PROGRAM 5: BREADTH FIRST SEARCH (BFS)

Aim

To write a Python program to implement Breadth First Search (BFS) traversal of a graph.

Objective

To traverse all the vertices of a graph level by level using the BFS algorithm.

Algorithm

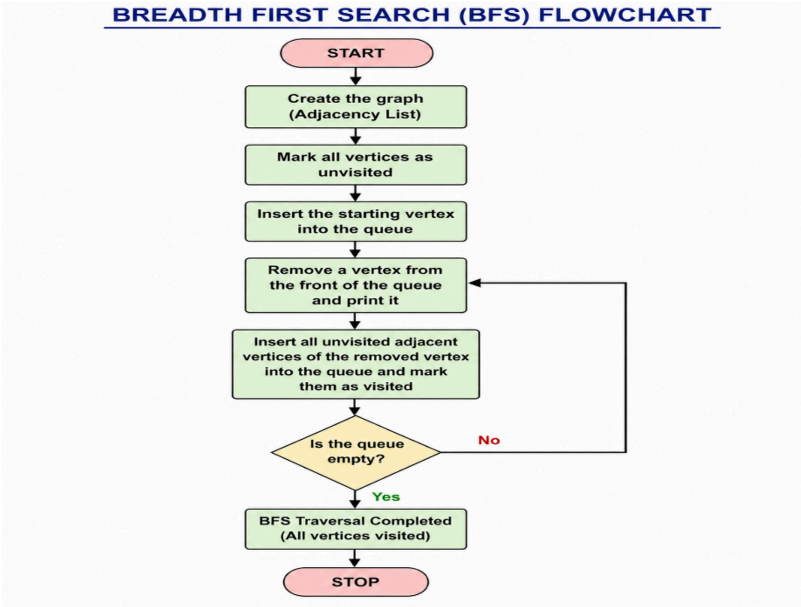
1. Start.
2. Create a graph using an adjacency list.
3. Mark all vertices as unvisited.
4. Insert the starting vertex into a queue.
5. Remove a vertex from the queue and print it.
6. Add all unvisited adjacent vertices to the queue.
7. Repeat until the queue becomes empty.
8. Stop.

Source Code

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
```

```
}

visited = []
queue = []

visited.append('A')
queue.append('A')

while queue:
    node = queue.pop(0)
    print(node, end=" ")

    for i in graph[node]:
        if i not in visited:
            visited.append(i)
            queue.append(i)
```

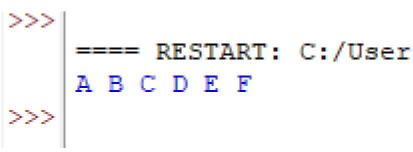
Flowchart



Sample Output

BFS Traversal:

A B C D E F



Result

Thus, the Python program to implement Breadth First Search (BFS) was executed successfully.